



EXERCISES OF PROBABILITY AND STATISTICS

Chapter 1

PROBABILITY.

1.1 RANDOM EXPERIMENTS AND METHODS OF COUNTING

Problem 1.1. (Bank Loan Processing)

A bank processes loan applications at one of three approval speeds depending on internal workload: Fast (f), Normal (n), Slow (s). Each approved application is either: Small loan (m) – under \$50,000; Large loan (l) – \$50,000 or more. Consider the experiment of observing one processed loan and recording the approval speed and loan size. An observation is a two-letter word. For example, a fast approval of a small loan is written as "fm".

- (a) What is the sample space of the experiment?
- (b) Let A_1 be the event "normal-speed approval." What are the outcomes in A_1 ?
- (c) Let A_2 be the event "small loan." What are the outcomes in A_2 ?
- (d) Let A_3 be the event "fast approval or slow approval." What are the outcomes in A_3 ?
- (e) Are A_1 , A_2 , and A_3 mutually exclusive?
- (f) Are A_1 , A_2 , and A_3 exhaustive?

Problem 1.2.

Let the sample space of the experiment consist of the measured resistances of two resistors. Give two examples of sets of events that are mutually exclusive and exhaustive.

Problem 1.3. (Market Performance of Two Firms).

An economic experiment observes the quarterly performance of two firms operating in the same industry:

- x = performance level of Firm A
- y = performance level of Firm B

Each firm's quarterly performance is classified as:

1 = Loss, 2 = Low profit, 3 = Moderate profit,
4 = High profit, 5 = Very high profit, 6 = Exceptional profit

Thus, each observation is an ordered pair (x, y) .

- (a) Describe the sample space S by listing the elements (x, y) .
- (b) List the elements corresponding to the event A that the total performance score is greater than 8.
- (c) List the elements corresponding to the event B that a low-profit quarter (2) occurs for either firm.
- (d) List the elements corresponding to the event C that Firm A achieves a performance greater than 4.
- (e) List the elements corresponding to the events $A \cap B; B \cap C; C \cap A$.
- (f) Construct a Venn diagram to illustrate the intersections and unions of the events A, B , and C .

Problem 1.4.

A train consists of four carriages. There are six passengers standing on the station platform.

- (a) How many ways are there to arrange six passengers in four carriages?
- (b) How many ways are there to arrange six passengers in four carriages such that the first carriage consists of 3 passengers, the second carriage consists of 2 passengers and the third carriage consists of 1 passenger?
- (c) How many ways are there to arrange six passengers in four carriages such that one carriage consists of 3 passengers, one carriage consists of 2 passengers and one carriage consists of 1 passenger?
- (d) How many ways are there to arrange six passengers in four carriages such that each carriage consists of at least one passenger?

 Problem 1.5.

On a V.League 1 team, the squad consists of:

- 18 outfield players (defenders, midfielders, forwards)
- 3 goalkeepers

For a match, the coach must select:

- 10 outfield players
- 1 goalkeeper

A starting lineup specifies:

- Which 10 outfield players are selected
- Which goalkeeper is selected
- The assignment of the 10 outfield players to 10 distinct positions in the tactical formation (e.g., LB, CB1, CB2, RB, CM1, CM2, LW, RW, ST1, ST2)

Assume:

- The goalkeeper position is fixed (only 1 position).
- Each tactical position is distinct.
- Only outfield players can occupy outfield positions.

How many possible starting lineups are there?

 Problem 1.6.

Four married couples have bought 8 seats in the same row for a concert. In how many different ways can they be seated

- (a) with no restrictions?
- (b) if each couple is to sit together?

 Problem 1.7.

- (a) In how many ways can 6 people be lined up to get on a bus?
- (b) If 3 specific persons, among 6, insist on following each other, how many ways are possible?
- (c) If 2 specific persons, among 6, refuse to follow each other, how many ways are possible?

1.2 PROBABILITY OF EVENTS AND PROBABILITY RULES.

Problem 1.8.

From a group of 3 freshmen, 4 sophomores, 4 juniors and 3 seniors a committee of size 4 is randomly selected. Find the probability that the committee will consist of

- (a) 1 from each class;
- (b) 2 sophomores and 2 juniors;
- (c) Only sophomores and juniors.

Problem 1.9.

Suppose that a school band contains 10 students from the freshman class, 20 students from the sophomore class, 30 students from the junior class, and 40 students from the senior class. If 15 students are selected at random from the band, what is the probability that at least one student from each of the four classes?

Problem 1.10.

A box contains 24 light bulbs of which four are defective. If one person selects 10 bulbs from the box in a random manner, and a second person then takes the remaining 14 bulbs, what is the probability that all 4 defective bulbs will be obtained by the same person?

Problem 1.11.

In a certain city, three newspapers A, B, and C are published. Suppose that 60 percent of the families in the city subscribe to newspaper A, 40 percent of the families subscribe to newspaper B, and 30 percent of the families subscribe to newspaper C. Suppose also that 20 percent of the families subscribe to both A and B, 10 percent subscribe to both A and C, 20 percent subscribe to both B and C, and 5 percent subscribe to all three newspapers A, B, and C. What percentage of the families in the city subscribe to at least one of the three newspapers?

Problem 1.12.

Consider two events A and B with $P(A) = 0.4$ and $P(B) = 0.7$. Determine the maximum and minimum possible values of $P(A \cap B)$ and the conditions under which each of these values is attained.

 Problem 1.13.

Suppose that A, B and C are three independent events such that $P(A) = \frac{1}{4}$, $P(B) = \frac{1}{3}$ and $P(C) = \frac{1}{2}$.

- (a) What is the probability that none of these three events will occur?
- (b) Determine the probability that exactly one of these three events will occur.

 Problem 1.14.

Suppose that four guests check their hats when they arrive at a restaurant, and that these hats are returned to them in a random order when they leave.

- (a) Determine the probability that no guest will receive the proper hat.
- (b) Determine the probability that at least 2 guests will receive the proper hat.

 Problem 1.15.

Mobile telephones perform hand-offs as they move from cell to cell. During a call, a telephone either performs zero hand-offs (H_0), one hand-off (H_1), or more than one hand-off (H_2). In addition, each call is either long (L), if it lasts more than three minutes, or brief (B). The following table describes the probabilities of the possible types of calls.

	H_0	H_1	H_2
L	0.1	0.1	0.2
B	0.4	0.1	0.1

- (a) What is the probability $P[H_0]$ that a phone makes no hand-offs?
- (b) What is the probability a call is brief?
- (c) What is the probability a call is long or there are at least two hand-offs?

 Problem 1.16.

A potential investor investigates the relationship between the performance of mutual funds and the university where the fund manager earned his/her MBA. Let A be the event that "a mutual fund outperformed the market" and let B be the event that "the fund manager graduated from top 20 MBA program". The joint probability table based on the data collected from 200 mutual funds for the investigation are given below:

	A	$\bar{A}$
B	0.11	0.29
$\bar{B}$	0.06	0.54

Select at random a mutual fund.

- What is the probability that the mutual fund outperformed the market?
- What is the probability that the mutual fund outperformed the market or the fund manager graduated from top 20 MBA program?
- What is the probability that the mutual fund outperformed the market given that the fund manager has graduated from top 20 MBA program?
- What is the probability that the mutual fund outperformed the market or the fund manager did not graduate from top 20 MBA program?
- Are the performance of mutual funds and the university where the fund manager earned his/her MBA independent?

 Problem 1.17.

A manufacturer of a flu vaccine is concerned about the quality of its flu serum. Batches of serum are processed by three different departments having rejection rates of 0.15, 0.12, and 0.09, respectively. The inspections by the three departments are sequential and independent.

- What is the probability that a batch of serum survives the first departmental inspection but is rejected by the second department?
- Given that a batch of serum has been rejected, what is the probability that it is rejected by the second department?

 Problem 1.18.

An aerospace company has submitted bids on two separate federal government defense contracts. The company president believes that there is a 40% probability of winning the first contract. If they win the first contract, the probability of winning the second is 70%. However, if they lose the first contract, the president thinks that the probability of winning the second contract decreases to 50%.

- What is the probability that they win both contracts?
- What is the probability that they lose both contracts?
- What is the probability that they win only one contract?

 Problem 1.19.

An investor believes that on a day when the Dow Jones Industrial Average (DJIA) increases, the probability that the NASDAQ also increases is 77%. If the investor believes that there is a 60% probability that the DJIA will increase tomorrow, what is the probability that the NASDAQ will increase as well?

 Problem 1.20.

A union's executive conducted a survey of its members to determine what the membership felt were the important issues to be resolved during upcoming negotiations with management. The results indicate that 74% of members felt that job security was an important issue, whereas 65% identified pension benefits as an important issue. Of those who felt that pension benefits were important, 60% also felt that job security was an important issue. One member is selected at random.

- What is the probability that he or she felt that both job security and pension benefits were important?
- What is the probability that the member felt that at least one of these two issues was important?

 Problem 1.21.

In an experiment, A , B , C , and D are events with probabilities $P[A] = \frac{1}{4}$, $P[B] = \frac{1}{8}$, $P[C] = \frac{5}{8}$, and $P[D] = \frac{3}{8}$. Furthermore, A and B are disjoint, while C and D are independent.

- Find $P[A \cap B]$, $P[A \cup B]$, $P[A \cap B^c]$, and $P[A \cup B^c]$.
- Are A and B independent?
- Find $P[C \cap D]$, $P[C \cap D^c]$, and $P[C^c \cap D^c]$.
- Are C^c and D^c independent?

 Problem 1.22.

In an experiment, A , B , C , and D are events with probabilities $P[A \cup B] = \frac{5}{8}$, $P[A] = \frac{3}{8}$, $P[C \cap D] = \frac{1}{3}$, $P[C] = \frac{1}{2}$. Furthermore, A and B are disjoint, while C and D are independent.

- Find $P[A \cap B]$, $P[B]$, $P[A \cap B^c]$, and $P[A \cup B^c]$.
- Are A and B independent?
- Find $P[D]$, $P[C \cap D^c]$, $P[C^c \cap D^c]$, $P[C|D]$.
- Find $P[C \cup D]$ and $P[C \cup D^c]$.
- Are C and D^c independent?

Problem 1.23.

Given the model of hand-offs and call lengths in Problem 1.15,

- (a) What is the probability that a brief call will have no hand-offs?
- (b) What is the probability that a call with one hand-off will be long?
- (c) What is the probability that a long call will have one or more hand-offs?

Problem 1.24.

You have a six-sided die that you roll once. Let R_i denote the event that the roll is i . Let G_j denote the event that the roll is greater than j . Let E denote the event that the roll of the die is even-numbered.

- (a) What is $P[R_3|G_1]$, the conditional probability that 3 is rolled given that the roll is greater than 1?
- (b) What is the conditional probability that 6 is rolled given that the roll is greater than 3?
- (c) What is $P[G_3|E]$, the conditional probability that the roll is greater than 3 given that the roll is even?
- (d) Given that the roll is greater than 3, what is the conditional probability that the roll is even?

Problem 1.25.

Two different suppliers, A and B, provide a manufacturer with the same part. All suppliers of this part are kept in a large bin. In the past, 5 percent of the parts supplied by A and 9 percent of the parts supplied by B have been defective. A supplies four times as many parts as B. Suppose you reach into the bin and select a part and find it is non-defective. What is the probability that it was supplied by A?

Problem 1.26.

Suppose that 30 percent of the bottles produced in a certain plant are defective. If a bottle is defective, the probability is 0.9 that an inspector will notice it and remove it from the filling line. If a bottle is not defective, the probability is 0.2 that the inspector will think that it is defective and remove it from the filling line.

- (a) If a bottle is removed from the filling line, what is the probability that it is defective?
- (b) If a customer buys a bottle that has not been removed from the filling line, what is the probability that it is defective?

 Problem 1.27.

Suppose that traffic engineers have coordinated the timing of two traffic lights to encourage a run of green lights. In particular, the timing was designed so that with probability 0.75 a driver will find the second light to have the same color as the first. Assuming the first light is equally likely to be red or green.

- (a) What is the probability that the second light is green?
- (b) What is the probability that you wait for at least one light?

 Problem 1.28.

A factory has three machines A, B, and C. Past records show that the machine A produced 40% of the items of output, the machine B produced 35% of the items of output, and machine C produced 25% of the items. Further 2% of the items produced by machine A were defective, 1.5% produced by machine B were defective, and 1% produced by machine C were defective.

- (a) If an item is drawn at random, what is the probability that it is defective?
- (b) An item is acceptable if it is not defective. What is the probability that an acceptable item comes from machine A?

 Problem 1.29.

A financial analyst estimates that the probability that the economy will experience a recession in the next 12 months is 25%. She also believes that if the economy encounters a recession, the probability that her mutual fund will increase in value is 20%. If there is no recession, the probability that the mutual fund will increase in value is 75%. Find the probability that the mutual fund's value will increase.

 Problem 1.30.

The Rapid Test is used to determine whether someone has HIV (the virus that causes AIDS). The false-positive and false-negative rates are 2.7% and 8%, respectively. A physician has just received the Rapid Test report that his patient tested positive. Before receiving the result, the physician assigned his patient to the low-risk group (defined on the basis of several variables) with only a 0.5% probability of having HIV. What is the probability that the patient actually has HIV?

Problem 1.31.

A customer-service supervisor regularly conducts a survey of customer satisfaction. The results of the latest survey indicate that 8% of customers were not satisfied with the service they received at their last visit to the store. Of those who are not satisfied, only 22% return to the store within a year. Of those who are satisfied, 64% return within a year. A customer has just entered the store. In response to your question, he informs you that it is less than 1 year since his last visit to the store. What is the probability that he was satisfied with the service he received?

Problem 1.32.

A telemarketer sells magazine subscriptions over the telephone. The probability of a busy signal or no answer is 65%. If the telemarketer does make contact, the probability of 0, 1, 2, or 3 magazine subscriptions is 0.5, 0.25, 0.2, and 0.05, respectively. Find the probability that in one call she sells no magazines.

Problem 1.33a.

There are 3 boxes of marbles: the first box contains 3 red marbles, 2 white marbles; the second box contains 2 red marbles, 2 white marbles; the third box has no marbles. Draw randomly 1 marble from the first box and 1 marble from the second box and put them in the third box. Then, from the third box, 1 marble is drawn at random. Given that the marble drawn from the third box is red, what is the probability that the marble drawn from the first box is red?

Problem 1.33b.

There are 2 boxes of marbles: the first box contains 3 red marbles, 2 white marbles; the second box contains 2 red marbles, 2 white marbles. Draw randomly 1 marble from the first box and put it in the second box. Then, from the second box, 1 marble is drawn at random and return it in the first box. Finally, 1 marble is drawn at random from the first box. Given that the marble drawn at the third step is red, what is the probability that the marble drawn at the first step is red?

 Problem 1.34.

An online retailer tracks whether a customer purchases a product during each of 5 independent promotional emails. For each email:

- 1 = customer makes a purchase
- 0 = customer does not purchase

Assume that for each email: $P(0) = 0.8$, $P(1) = 0.2$ and responses to different emails are independent. An example outcome is 01010, meaning the customer purchased only on the second and fourth emails.

- (a) What is the probability of the sequence 00111?
- (b) What is the probability that the customer makes exactly three purchases?

 Problem 1.35.

Suppose each day that you drive to work a traffic light that you encounter is either green with probability $\frac{7}{16}$, red with probability $\frac{7}{16}$, or yellow with probability $\frac{1}{8}$, independent of the status of the light on any other day. If over the course of five days, G , Y , and R denote the number of times the light is found to be green, yellow, or red, respectively, what is the probability that $P[G = 2, Y = 1, R = 2]$? Also, what is the probability $P[G = R]$?

Chapter 2

RANDOM VARIABLES AND PROBABILITY DISTRIBUTIONS.

2.1 DISCRETE RANDOM VARIABLES.

Problem 2.1.

A civil engineer is studying a left-turn lane that is long enough to hold 7 cars. Let X be the number of cars in the lane at the end of a randomly chosen red light. The engineer believes that the probability that $X = x$ is proportional to $(x + 1)(8 - x)$ for $x = 0, 1, \dots, 7$.

- Find the probability mass function of X .
- Find the probability that X will be at least 5.
- Find the cumulative distribution function of X .
- Find the measures of central location of X : the expectation, the median and the mode.
- Find the measures of variability of X : the variance and the standard deviation.
- Find the probability distribution table of $Y = 3X^2 + 1$.

Problem 2.2.

A midterm test has 4 multiple choice questions with four choices with one correct answer each. If you just randomly guess on each of the 4 questions, what is the probability that you get exactly 2 questions correct? Assume that you answer all and you will get (+5) points for 1 question correct, (-2) points for 1 question wrong.

- Find the probability mass function of X .
- Find the probability that X will be positive.
- Find the cumulative distribution function of X .
- Find the measures of central location of X : the expectation, the median and the mode.
- Find the measures of variability of X : the variance and the standard deviation.

 Problem 2.3.

The random variable N has the following pmf $f_N(n) = \begin{cases} c \left(\frac{1}{2}\right)^2, & n = 0, 1, 2 \\ 0, & \text{otherwise} \end{cases}$

- (a) What is the value of the constant c ?
- (b) What is $P[N \leq 1]$?
- (c) What is the cdf of N ?

 Problem 2.4.

There are two boxes, the first box consists of 7 red balls and 3 white balls, the second box consists of 5 red balls and 2 white balls. Draw randomly 2 balls from the first box to second one, then continue to draw randomly 2 balls from the second one. Let X be the number of white balls out of these 2 balls.

- (a) Find the probability distribution table of X .
- (b) Find the cumulative distribution function of X .
- (c) Find the measures of central location of X : the expectation, the median and the mode.
- (d) Find the measures of variability of X : the variance and the standard deviation.

 Problem 2.5.

When a conventional paging system transmits a message, the probability that the message will be received by the pager it is sent to is p . To be confident that a message is received at least once, a system transmits the message n times.

- (a) Assuming all transmissions are independent, what is the pmf of K , the number of times the pager receives the same message?
- (b) Assume $p = 0.8$. What is the minimum value of n that produces a probability of 0.95 of receiving the message at least once?

 Problem 2.6.

When a two-way paging system transmits a message, the probability that the message will be received by the pager it is sent to is p . When the pager receives the message, it transmits an acknowledgment signal (ACK) to the paging system. If the paging system does not receive the ACK, it sends the message again.

- (a) What is the pmf of N , the number of times the system sends the same message?
- (b) The paging company wants to limit the number of times it has to send the same message. It has a goal of $P[N \leq 3] \geq 0.95$. What is the minimum value of p necessary to achieve the goal?

 Problem 2.7.

The random variable X has the following cdf: $F_X(x) = \begin{cases} 0, & x \leq -1, \\ 0.2, & -1 < x \leq 0, \\ 0.7, & 0 < x \leq 1, \\ 1, & x > 1. \end{cases}$

- Draw a graph of the cdf.
- Write $f_X(x)$, the pmf of X . Be sure to write the value of $f_X(x)$ for all x from $-\infty$ to $+\infty$.
- Write the probability distribution table of X .

 Problem 2.8.

The random variable X has the following cdf: $F_X(x) = \begin{cases} 0, & x \leq -3, \\ c, & -3 < x \leq 5, \\ 0.8, & 5 < x \leq 7, \\ 1, & x > 7. \end{cases}$

- Find the constant c given that $P(X = -3) = P(X = 5)$.
- Draw a graph of the cdf.
- Write $f_X(x)$, the pmf of X .
- Write the probability distribution table of X .

 Problem 2.9.

A shipment of 7 television sets contains 2 defective sets. A hotel makes a random purchase of 3 of the sets. Let X be the number of defective sets purchased by the hotel.

- Find the probability distribution of X .
- Find the cumulative distribution function of the random variable X .
- Find $P(X = 1)$ and $P(0 < X \leq 2)$.

 Problem 2.10.

Voice calls cost 20 cents each and data calls cost 30 cents each. X is the cost of one telephone call. The probability that a call is a voice call is $P[V] = 0.6$. The probability of a data call is $P[D] = 0.4$.

- Find the pmf of X .
- What is $E[X]$, the expected value of X ?

 Problem 2.11.

Given the random variable X in Problem 2.7, let $V = g(X) = |X|$.

- (a) Find the pmf $f_V(v)$ of V .
- (b) Find the cdf $F_V(v)$ of V .
- (c) Find $E[V]$.

 Problem 2.12.

In a certain lottery game, the chance of getting a winning ticket is exactly one in a thousand. Suppose a person buys one ticket each day (except on the leap year day February 29) over a period of fifty years. What is the expected number $E[T]$ of winning tickets in fifty years? If each winning ticket is worth \$1000, what is the expected amount $E[R]$ collected on these winning tickets? Lastly, if each ticket costs \$2, what is your expected net profit $E[Q]$?

 Problem 2.13.

Show that the variance of $Y = aX + b$ is $\text{Var}[Y] = a^2 \text{Var}[X]$

 Problem 2.14.

Given a random variable X with mean μ_X and variance σ_X^2 , find the mean and variance of the standardized random variable $Y = \frac{X - \mu_X}{\sigma_X}$.

2.2 CONTINUOUS RANDOM VARIABLES.

 Problem 2.15.

Let X be the maximum price (in hundreds of dollars) a consumer is willing to pay for a product. Suppose that X has the following probability density function $f_X(x) =$

$$\begin{cases} cx, & 0 \leq x \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Find the constant c .
- (b) Find $P[0 \leq X \leq 1]$.
- (c) Find the cumulative distribution function of X .
- (d) Find the measures of central location of X : the expectation and the median.
- (e) Find the measures of variability of X : the variance and the standard deviation.
- (f) Let a firm set a price $p \in [0, 2]$, and define a new variable Y as the consumer surplus variable (measures how much benefit a buyer gets beyond the price): $Y = \max(X - p, 0)$. Find the cdf of Y .

 Problem 2.16.

Let the random variable X measure how much a consumer's valuation deviates from an average level. Suppose that the cumulative distribution function of random variable X is

$$F_X(x) = \begin{cases} 0, & x \leq -1, \\ \frac{x+1}{2}, & -1 < x \leq 1, \\ 1, & x > 1 \end{cases}$$

- (a) What is $P[X \leq 1/2]$?
- (b) What is the value of a such that $P[X < a] = 0.8$?
- (c) Find the measures of central location of X : the expectation and the median.
- (d) Find the measures of variability of X : the variance and the standard deviation.
- (e) Find the cdf of $Y = e^X$ and compute the expectation of Y by 2 methods.

 Problem 2.17.

Let V be the net payoff (profit or benefit) from an economic activity with favorable prospects. Suppose that the cumulative distribution function of V is

$$F_V(v) = \begin{cases} 0, & v \leq -5, \\ c(v+5)^2, & -5 < v \leq 7, \\ 1, & v > 7 \end{cases}$$

- (a) What is c ?
- (c) What is $P[-3 \leq v < 0]$?
- (d) What is the value of a such that $P[V \geq a] = 2/3$?
- (c) Find the expectation and the variance of X .
- (e) Let U be the utility from profit and define by $U = \sqrt{V+5}$. Find the expectation of U by 2 methods.

 Problem 2.18.

Continuous random variable X has the following pdf $f_X(x) = \begin{cases} 1/4, & -1 \leq x \leq 3, \\ 0, & \text{otherwise.} \end{cases}$.

Define the random variable Y by $Y = h(X) = X^2$.

- (a) Find $E[X]$ and $V[X]$.
- (b) Find $h(E[X])$ and $E[h(X)]$.
- (c) Find $V[Y]$.

 Problem 2.19.

A continuous random variable X has the following pdf $f_X(x) = Ce^{-|x|}$ for all $x \in \mathbb{R}$.

- (a) Find the normalizing constant C .
- (b) Find the cdf of X .
- (c) Find the cdf and pdf of $Y = X^2$.
- (d) Find $E(X)$; $V(X)$.
- (e) Find the probability that out of 5 independent trials of observing the value of X , there are exactly 3 times that X takes value in the interval $[2; 3]$.

 Problem 2.20.

The cumulative distribution function of the continuous random variable X is

$$F_X(x) = \begin{cases} 0, & x \leq 0, \\ \frac{1}{2} - k \cos x, & 0 < x \leq \pi, \\ 1, & x > \pi. \end{cases}$$

- (a) What is k ?
- (b) What is $P\left[0 < x < \frac{\pi}{2}\right]$?
- (c) What is $E[X]$?

 Problem 2.21.

The cumulative distribution function of the continuous random variable X is

$$F(x) = \begin{cases} 0, & x \leq -a, \\ A + B \arcsin \frac{x}{a}, & -a < x < a, \\ 1, & x \geq a. \end{cases}$$

- (a) What are A and B ?
- (b) What is the pdf $f_X(x)$?

 Problem 2.22.

The cumulative distribution function of the continuous random variable X is

$$F(x) = a + b \arctan x, \quad (-\infty < x < +\infty)$$

- (a) What are a and b ?
- (b) What is pdf $f_X(x)$?
- (c) What is $P[-1 < X < 1]$?

 Problem 2.23.

The continuous random variable X has probability density function

$$f(x) = \begin{cases} k \sin 3x, & x \in \left(0; \frac{\pi}{3}\right), \\ 0, & x \notin \left(0; \frac{\pi}{3}\right). \end{cases}$$

Use the pdf to find

- (a) the constant k .
- (b) cdf $F_X(x)$.
- (c) $P\left[\frac{\pi}{6} \leq X < \frac{\pi}{3}\right]$.

 Problem 2.24.

The continuous random variable X has a pdf $f(x) = \frac{c}{e^x + e^{-x}}$. What is $E[X]$?

2.3 SOME IMPORTANT PROBABILITY DISTRIBUTIONS.

 Problem 2.25.

A company sells a product to consumers whose willingness to pay (WTP, denoted by W) is uniformly distributed between 10 and 50 (in dollars). The firm sets a price p .

- (a) What is the probability that a randomly selected consumer will buy the product (i.e., $W \geq p$)?
- (b) Find the firm's expected revenue as a function of price p (Revenue = price*probability of purchase:). What price maximizes expected revenue?

 Problem 2.26.

X is a continuous uniform random variable with expected value $\mu_X = 7$ and variance $\text{Var}[X] = 3$. What is the pdf of X ?

 Problem 2.27.

X is uniform random variable with parameters 0 and 1. Find a function $g(x)$ such that the pdf of $Y = g(X)$ is $f_Y(y) = \begin{cases} 3y^2, & 0 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$

 Problem 2.28.

Let X have a binomial pmf $f_X(x) = C_4^x \left(\frac{1}{2}\right)^4$, for $x \in \{0, 1, 2, 3, 4\}$.

- (a) Find the standard deviation of the random variable X .
- (b) What is $P[\mu_X - \sigma_X \leq X \leq \mu_X + \sigma_X]$, the probability that X is within one standard deviation of the expected value?

 Problem 2.29.

A large market has 20 potential firms considering entry. Each firm independently enters the market with probability $p = 0.3$. Let X be the number of firms entered the market out of these 20 firms.

- (a) What is the probability that there are 5 firms entered the market.
- (b) Find the mean, the variance and the mode of X .

 Problem 2.30.

A firm launches a new product. Each consumer independently buys the product with probability $p = 0.25$. The firm targets $n = 20$ potential buyers.

- (a) Compute the probability that at least 8 consumers purchase the product by using a normal approximation.
- (b) The firm considers a marketing campaign that increases purchase probability to $p = 0.35$, but costs 50 USD. Each unit sold yields 10 USD profit. Compute expected profit before and after the campaign. Should the firm invest in marketing?

 Problem 2.31.

When a conventional paging system transmits a message, the probability that the message will be received by the pager it is sent to is p . To be confident that a message is received at least once, a system transmits the message n times.

- (a) Assuming all transmissions are independent, what is the pmf of K , the number of times the pager receives the same message?
- (b) Assume $p = 0.8$. What is the minimum value of n that produces a probability of 0.95 of receiving the message at least once?

 Problem 2.32.

Customers arrive at a small bank according to a Poisson process with rate $\lambda = 6$ customers per hour.

- What is the probability that exactly 4 customers arrive in one hour?
- What is the probability that at exactly 8 customers arrive in two hours?
- The bank can serve at most 7 customers per hour. If more than 7 arrive, the excess customers leave (lost revenue). Compute the probability of lost customers and the expected number of lost customers.
- The bank considers hiring an additional teller, increasing capacity to 10 customers/hour at a cost of 30 USD/hour. Each customer generates 8 USD in profit. Should the bank expand capacity?

 Problem 2.33.

A car rental garage observes that the number of customers arriving to rent cars on a Saturday follows a Poisson distribution with parameter $\lambda = 2$. Assume that the garage has 4 cars available for rental.

- Find the probability that all 4 cars are rented on Saturday.
- Find the probability that the garage cannot meet demand (i.e., there is a shortage of cars) on Saturday.
- Given that there were at least 2 customers arriving to rent cars, what is the probability that there were exactly 4 customers arriving to rent cars.
- On average, how many cars are rented on a Saturday?

 Problem 2.34.

Wages in a labor market are normally distributed with a mean of 50 000 USD and a standard deviation of 10 000 USD.

- What proportion of workers earn more than 65,000 USD?
- The government defines “high income” as the top 10%. What income level corresponds to this threshold?
- Suppose a tax applies only to those earning above 70 000 USD. What fraction of workers pay this tax?

 Problem 2.35.

Demand for a product is normally distributed with a mean of 100 units and a variance of 20^2 (units²).

- What is the probability demand exceeds 130 units?
- If the firm produces 110 units, what is the probability of stockout?
- Compute the expected number of unsatisfied units.
- If each unit yields 5 USD profit and shortage costs 8 USD per unit, compute expected profit.

Problem 2.36.

The duration (in months) of unemployment for an individual follows an exponential distribution: $T \sim \text{Exponential}(\lambda = 0.2)$.

- (a) What is the probability that an individual remains unemployed for more than 6 months?
- (b) What is the probability that an individual who has already been unemployed for 4 months will remain unemployed for at least 3 more months?
- (c) Compute the expected duration of unemployment.
- (d) The government introduces a policy that reduces the expected unemployment duration to 4 months. What is the new value of λ ? What is the probability that unemployment exceeds 6 months under the new policy?

Problem 2.37.

A four engine plane can fly if at least two engines work.

- (a) If the engines operate independently and each malfunctions with probability q , what is the probability that the plane will fly safely?
- (b) A two engine plane can fly if at least one engine works and if an engine malfunctions with probability q , what is the probability that plane will fly safely?
- (c) Which plane is the safest?

Problem 2.38.

Let X be an exponential random variable with parameter and define $Y = [X]$, the largest integer in X , (ie. $[x] = 0$ for $0 \leq x < 1$, $[x] = 1$ for $1 \leq x < 2$ etc.)

- (a) Find the probability function for Y .
- (b) Find $E(Y)$.
- (c) Find the distribution function of Y .
- (d) Let Y represent the number of periods that a machine is in use before failure. What is the probability that the machine is still working at the end of 10th period given that it does not fail before 6th period?

Chapter 3

VECTORS OF RANDOM VARIABLES

3.1 PAIRS OF DISCRETE RANDOM VARIABLES.



Problem 3.1.

Consider a bank analyzing a portfolio of loans. Let X be the credit rating category of a borrower ($X = 1$: low-risk borrower and $X = 2$: high-risk borrower). Let Y be the loss severity level (if adverse event occurs; $Y = 1$: low loss, $Y = 2$: medium loss and $Y = 3$: high loss). Suppose that X and Y have the joint PDT as follows

$X \setminus Y$	1	2	3
1	0.12	0.15	0.03
2	0.28	0.35	0.07

- Find the marginal probability distribution (PDT) of X and of Y : the distribution of X helps the bank understand its risk composition (how many risky borrowers) and the distribution of Y is used in expected loss calculations and capital planning.
- Let Z be the risk-weighted loss index and defined by $Z = XY$. Find the PDT of Z : this distribution is used in stress testing and portfolio risk ranking and the banks can classify loans into risk buckets.
- Find the conditional probability distribution of X given that $Y = 1$: given that the loss is low, what type of borrowers caused it? It identifies whether low losses come mostly from safe or risky clients and helps to detect the hidden risks and misclassification in credit scoring.
- Find the conditional probability distribution of Y given that $X = 1$: for low-risk borrowers, what is the distribution of losses? It measures loss behavior within a rating class.
- Are X and Y independent?
- Find the covariance matrix of (X, Y) .
- Find the coefficient of correlation between X and Y .
- Are X and Y related? Interpret the result.

Problem 3.2.

Let X and Y be demands (in units) for two substitute products with a joint pmf given as follows

$$P(X = x, Y = y) = c(x + y), \quad x, y \in \{0, 1, 2\}.$$

- Find the constant c .
- Compute the marginal distributions.
- Are X and Y independent?
- Interpret the sign of $\text{Cov}(X, Y)$ economically.

Problem 3.3.

Random variables X and Y have the joint PDT as the following

X	Y	-1	0	1
-1		$\frac{4}{15}$	$\frac{1}{15}$	$\frac{4}{15}$
0		$\frac{1}{15}$	$\frac{2}{15}$	$\frac{1}{15}$
1		0	$\frac{2}{15}$	0

- Find the marginal probability distribution (PDT) of X and of Y .
- Find the PDT of Z , where $Z = X + Y$.
- Find the conditional probability distribution of X given that $Y = -1$.
- Find the conditional probability distribution of Y given that $X = 1$.
- Are X and Y independent?
- Find the covariance of (X, Y) .
- Find the coefficient of correlation between X and Y .
- Are X and Y related?

Problem 3.4.

Vector (X, Y) has the following joint PMF

$$f(x, y) = P(X = x, Y = y) = \begin{cases} cxy, & x = 1, 2, 4; y = 1, 3, \\ 0, & \text{otherwise.} \end{cases}$$

- What is the value of the constant c ?
- What is $P[Y < X]$?
- Find the marginal probability distribution (PDT) of X and of Y .
- Find the conditional probability distribution of X given that $Y = 3$.
- Find the covariance matrix of (X, Y) .
- Find the coefficient of correlation between X and Y .
- Are X and Y related?

3.2 PAIRS OF CONTINUOUS RANDOM VARIABLES.

 Problem 3.5.

Let X be return of a stock and Y be the market return. Assume that X and Y have the joint PDF

$$f(x, y) = \begin{cases} c(1 + xy), & -1 \leq x \leq 1, -1 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) What is the value of the constant c ?
- (b) What is $P[X \leq Y]$?
- (c) Find the marginal CDF of X and of Y .
- (d) Find the conditional pdf of X given that $Y = 1/2$.
- (e) Find the conditional pdf of Y given that $X = 1/3$.
- (f) Find the covariance of (X, Y) . Interpret the sign of covariance in terms of systematic risk.
- (g) Find the coefficient of correlation between X and Y .
- (h) Are X and Y related? independent?

 Problem 3.6.

Random variables X and Y have the joint PDF

$$f(x, y) = \begin{cases} cy, & 0 \leq y \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Draw the region of nonzero probability.
- (b) What is the value of the constant c ?
- (c) Find the marginal CDF of X and of Y .
- (d) Find the conditional pdf of X given that $Y = 1/2$.
- (e) Find the covariance of (X, Y) .
- (f) Find the coefficient of correlation between X and Y .
- (g) Are X and Y related? independent?

Problem 3.7.

Random variables X and Y have joint PDF

$$f(x, y) = \begin{cases} cxy^2, & 0 \leq x \leq 1, 0 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

- What is the value of the constant c ?
- Find $P[X > Y]$ and $P[Y < X^2]$.
- Find $P\left[\min(X, Y) \leq \frac{1}{2}\right]$.
- Find the marginal PDF of X and of Y .
- Find the conditional pdf of X given that $Y = 1/2$.
- Find the conditional pdf of Y given that $X = 1/3$.
- Find the covariance of (X, Y) .
- Find the coefficient of correlation between X and Y .
- Are X and Y related? independent?

Problem 3.8.

Let the income X and the consumption Y have joint pdf:

$$f(x, y) = \begin{cases} cye^{-x}, & 0 < y < x < \infty, \\ 0, & \text{otherwise.} \end{cases}$$

- What is the value of the constant c ?
- Compute $E[Y|X = x]$. Interpret $E[Y|X = x]$ in terms of marginal propensity to consume.
- Find the covariance of (X, Y) .
- Are X and Y related? independent?